N+ Network Security

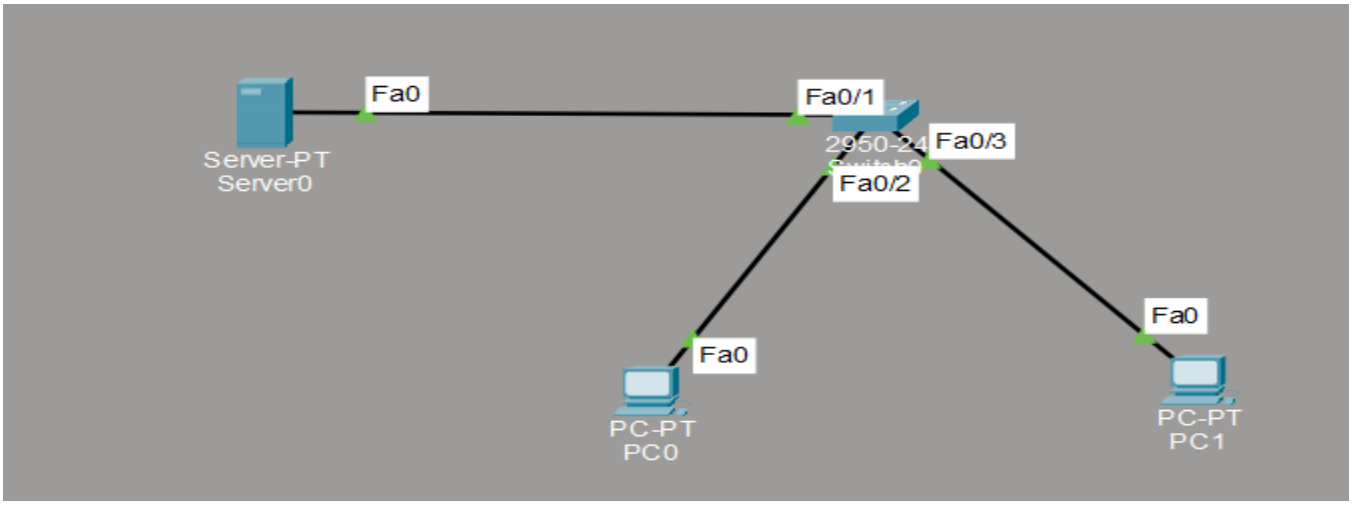
10/31/2026

Assignment-10



Prepare By
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1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.



DHCP Server Configuration in Cisco Packet Tracer

I created a simple network using one DHCP server, one switch, and two client PCs.

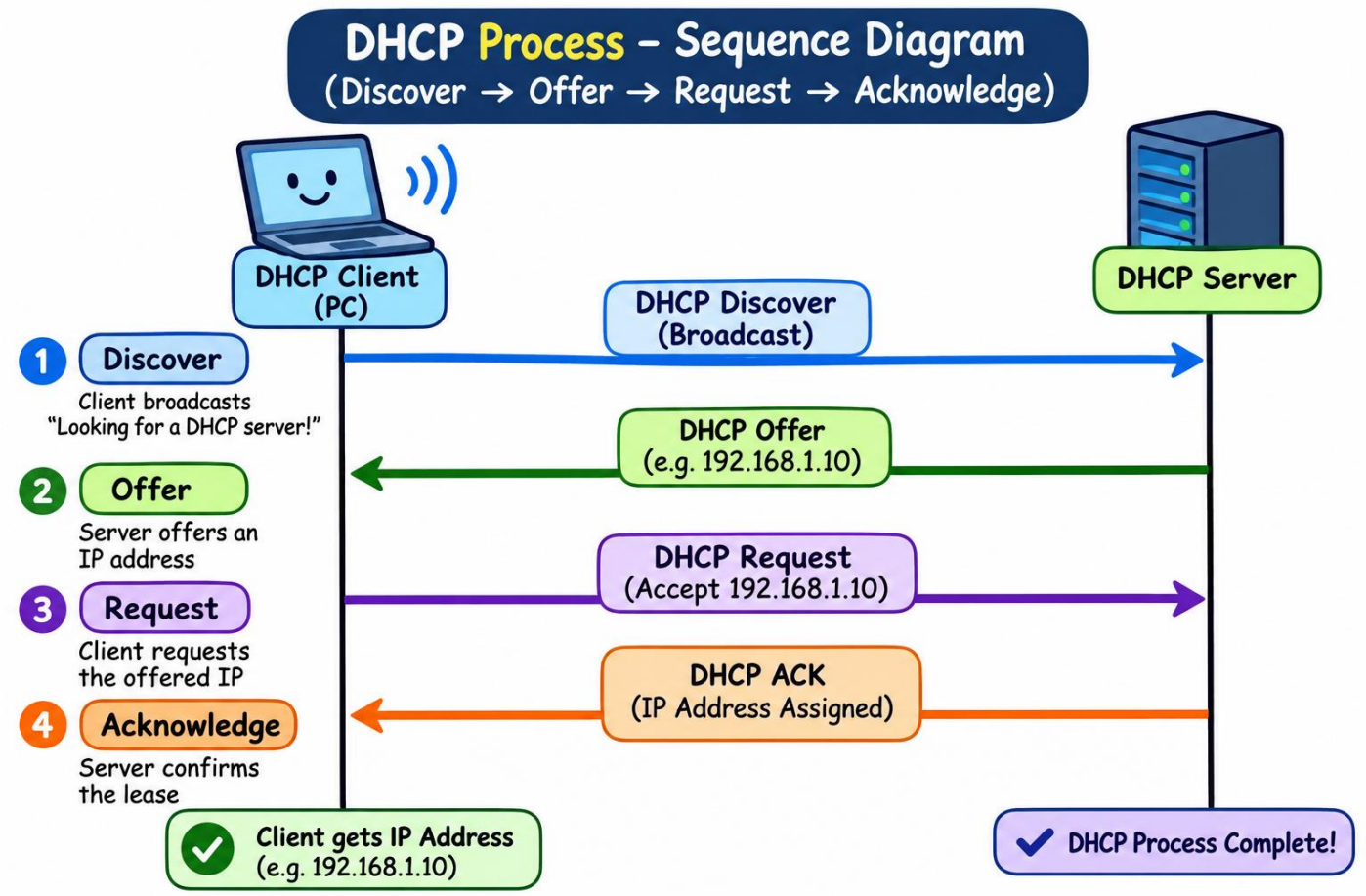
1. Connected the DHCP server and both PCs to the switch using Copper Straight-Through cables.
2. Assigned a static IP address 192.168.1.2 to the DHCP server.
3. Enabled the DHCP service on the server.
4. Created a DHCP pool with the following settings:
    - Start IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
5. Selected DHCP in the IP Configuration settings of both PCs.

Verification:

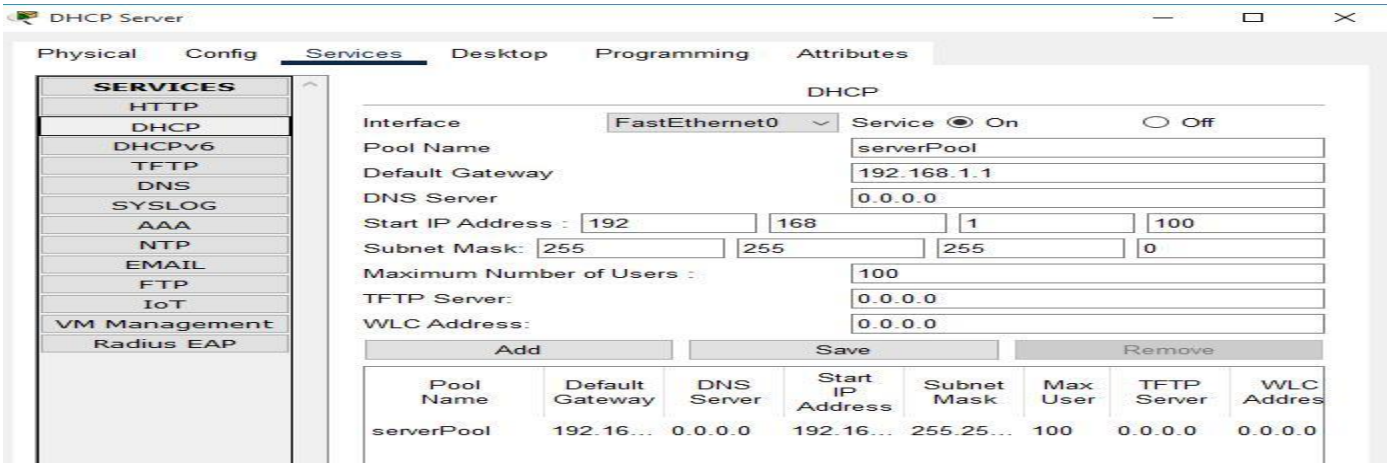
Device	IP Address
PC0	192.168.1.10
PC1	192.168.1.11

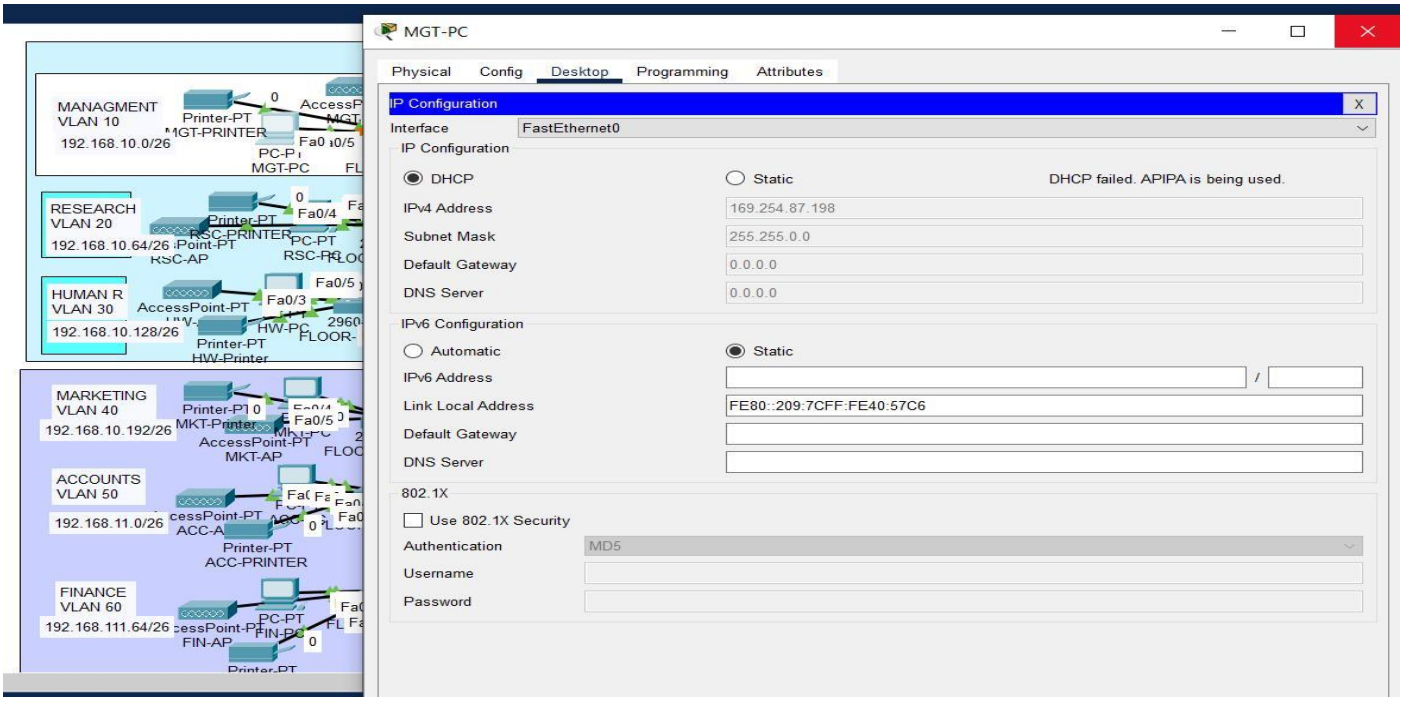
Result: Both PCs received different IP addresses automatically from the DHCP server.

2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.Hint: Use arrows to show the direction of each message between client and server.



3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.





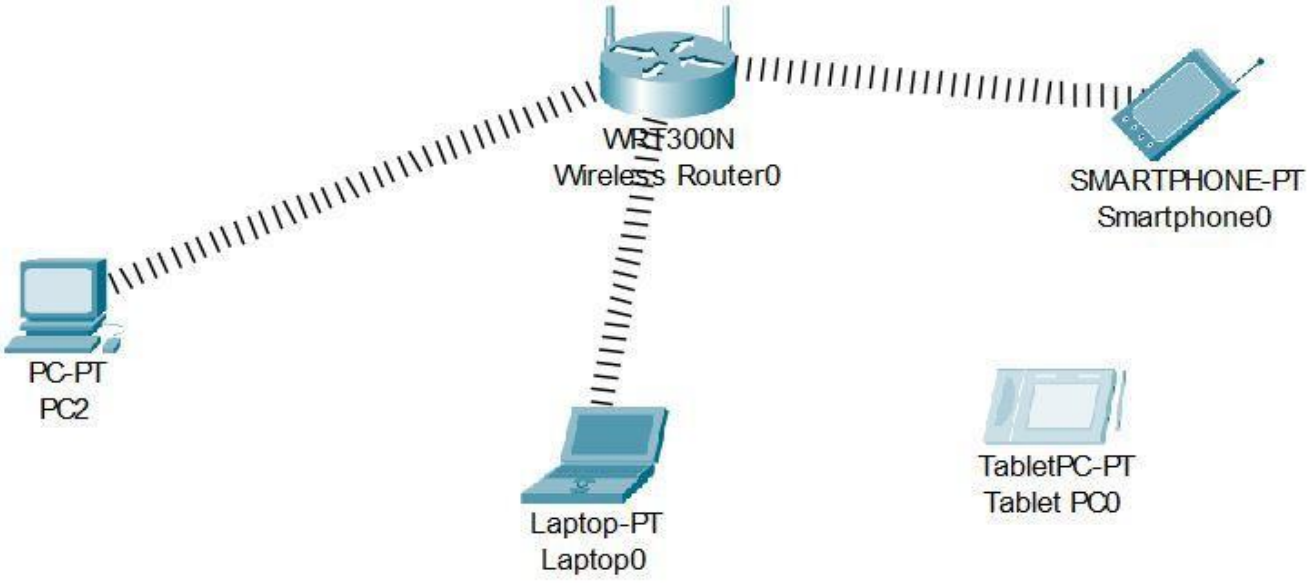
DHCP Pool Limitation Test

I configured the DHCP server with only one available IP address and connected two PCs.

- PC0 successfully received IP address 192.168.1.10.
- PC1 failed to get an IP address because the DHCP pool had no free IP addresses.

Conclusion: Only one PC succeeds because the DHCP server has only one available IP address in its pool.

4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.



I configured a DHCP server in Cisco Packet Tracer for a Zomato-style restaurant WiFi network. I connected a laptop and a generic PC (smartphone) to the WiFi network.

Both devices received IP addresses automatically through DHCP.

- Laptop IP: 192.168.1.100
- Smartphone IP: 192.168.1.101

Why is DHCP useful?

DHCP automatically assigns IP addresses to connected devices. It saves time and avoids manual IP configuration for restaurant customers.